
DSC 140A - Homework 05

Homework 05 Instructions. You may collaborate with whomever, including AI. Each student must submit their own homework even if they worked with others. We will not be grading for correctness, but we will be grading for completeness. Please turn out a single PDF file that contains your solutions to all of the problems. We recommend using $\text{\LaTeX}$ with our template. You can use markdown but make sure each problem is clearly labeled. For coding problems, we recommend using Jupyter notebooks, convert them into PDF, and then merge the PDF with your math problems solutions generated in $\text{\LaTeX}$.

Problem 1.

In this problem, you will derive the solution to the ridge regression optimization problem.

Recall that the ridge regression regularized risk function is

$$\tilde{R}(\vec{w}) = \frac{1}{n} \sum_{i=1}^n (\vec{\phi}(\vec{x}^{(i)}) \cdot \vec{w} - y_i)^2 + \lambda \|\vec{w}\|^2.$$

Here, $\vec{\phi}(\vec{x})$ is a feature map. Also recall that this risk function can be equivalently written in matrix-vector form as

$$\tilde{R}(\vec{w}) = \frac{1}{n} \|\Phi \vec{w} - \vec{y}\|^2 + \lambda \|\vec{w}\|^2,$$

where Φ is the design matrix; its i th row is $\vec{\phi}(\vec{x}^{(i)})$.

- a) Show that $\tilde{R}(\vec{w}) = \frac{1}{n} (\vec{w}^T \Phi^T \Phi \vec{w} - 2\vec{w}^T \Phi^T \vec{y} + \vec{y}^T \vec{y}) + \lambda \vec{w}^T \vec{w}$.

Solution: Using the fact that, for any vector $\vec{u}$, $\|\vec{u}\|^2 = \vec{u}^T \vec{u}$, we have:

$$\begin{aligned} \tilde{R}(\vec{w}) &= \frac{1}{n} \|\Phi \vec{w} - \vec{y}\|^2 + \lambda \|\vec{w}\|^2 \\ &= \frac{1}{n} (\Phi \vec{w} - \vec{y})^T (\Phi \vec{w} - \vec{y}) + \lambda \vec{w}^T \vec{w} \\ &= \frac{1}{n} (\vec{w}^T \Phi^T - \vec{y}^T) (\Phi \vec{w} - \vec{y}) + \lambda \vec{w}^T \vec{w} \\ &= \frac{1}{n} (\vec{w}^T \Phi^T \Phi \vec{w} - \vec{w}^T \Phi^T \vec{y} - \vec{y}^T \Phi \vec{w} + \vec{y}^T \vec{y}) + \lambda \vec{w}^T \vec{w} \end{aligned}$$

$\vec{y}^T \Phi \vec{w}$ is a scalar, and any scalar transposed is just itself. So $(\vec{y}^T \Phi \vec{w})^T = \vec{w}^T \Phi^T \vec{y}$, and we have:

$$= \frac{1}{n} (\vec{w}^T \Phi^T \Phi \vec{w} - 2\vec{w}^T \Phi^T \vec{y} + \vec{y}^T \vec{y}) + \lambda \vec{w}^T \vec{w}$$

- b) So far this quarter, we have seen a few vector calculus identities. For example, we know that $\frac{d}{d\vec{w}} (\vec{w}^T \vec{w}) = 2\vec{w}$.

Using these identities, show that

$$\frac{d}{d\vec{w}} \tilde{R}(\vec{w}) = \frac{1}{n} (2\Phi^T \Phi \vec{w} - 2\Phi^T \vec{y}) + 2\lambda \vec{w}.$$

Solution:

$$\begin{aligned} \frac{d}{d\vec{w}} \left[\frac{1}{n} (\vec{w}^T \Phi^T \Phi \vec{w} - 2\vec{w}^T \Phi^T \vec{y} - \vec{y}^T \vec{y}) + \lambda \vec{w}^T \vec{w} \right] \\ = \left[\frac{1}{n} \left(\frac{d}{d\vec{w}} \vec{w}^T \Phi^T \Phi \vec{w} - \frac{d}{d\vec{w}} 2\vec{w}^T \Phi^T \vec{y} - \frac{d}{d\vec{w}} \vec{y}^T \vec{y} \right) + \frac{d}{d\vec{w}} \lambda \vec{w}^T \vec{w} \right] \\ = \frac{1}{n} (2\Phi^T \Phi \vec{w} - \Phi^T \vec{y} - \vec{0}) + 2\lambda \vec{w} \end{aligned}$$

- c) Show that the minimizer of $\tilde{R}(\vec{w})$ is $\vec{w}^* = (\Phi^T \Phi + n\lambda I)^{-1} \Phi^T \vec{y}$.

Solution: Setting the gradient to zero and solving for $\vec{w}$, we have:

$$\begin{aligned} \frac{1}{n} (2\Phi^T \Phi \vec{w} - 2\Phi^T \vec{y}) + 2\lambda \vec{w} &= \vec{0} && \text{(multiply both sides by } n) \\ \Rightarrow 2\Phi^T \Phi \vec{w} - 2\Phi^T \vec{y} + 2\lambda n \vec{w} &= \vec{0} && \text{(grouping terms with } \vec{w}) \\ \Rightarrow 2\Phi^T \Phi \vec{w} + 2\lambda n \vec{w} &= 2\Phi^T \vec{y} && \text{(dividing both sides by 2)} \\ \Rightarrow \Phi^T \Phi \vec{w} + \lambda n \vec{w} &= \Phi^T \vec{y} && \text{(factoring out } \vec{w}) \\ \Rightarrow (\Phi^T \Phi + n\lambda I) \vec{w} &= \Phi^T \vec{y} && \text{(solving for } \vec{w}) \\ \Rightarrow \vec{w} &= (\Phi^T \Phi + n\lambda I)^{-1} \Phi^T \vec{y} \end{aligned}$$

Problem 2.

Suppose you are modeling the relationship between a single input x and a target y . You suspect the true relationship is quadratic, so you decide to use the feature map $\vec{\phi}(x) = (x, x^2)^T$. You have the following training data:

$x^{(i)}$	y_i
0	1
1	2
2	5

- a) Compute $\vec{\phi}(x^{(i)})$ for each training point and construct the augmented design matrix Φ .

Solution:

$$\Phi = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{pmatrix}$$

- b) The least squares solution in feature space is $\vec{w}^* = (1, 0, 1)^T$. Write the prediction function $H(x)$ explicitly as a function of x alone — not in terms of ϕ . Is this function linear in x ?

Solution: $H(x) = 1 + x^2$. This function is **not** linear in x .

- c) A new point arrives with $x = 3$. Using the two-step prediction procedure from lecture, compute $H(3)$.

Solution: First, map to feature space: $\vec{\phi}(3) = (3, 9)^T$. Then:

$$H(3) = \vec{w}^* \cdot \text{Aug}(\vec{\phi}(3)) = \vec{w}^* \cdot (1, 3, 9)^T = 1(1) + 0(3) + 1(9) = 10$$

- d) A classmate argues: “Since we train a linear model in feature space, the prediction function must also appear linear when plotted against the original input x .” Is this correct? Explain in one to two sentences.

Solution: Incorrect. The model is linear in the *mapped features*, but substituting $\vec{\phi}$ back in terms of x produces a non-linear function in input space — in this case, a parabola.

Problem 3.

In this problem, you will explore the geometry of the Hard-SVM and how scaling and feasibility affect the choice of classifier. Recall that the Hard-SVM optimization problem is

$$\vec{w}^* = \arg \min_{\vec{w}} \|\vec{w}\|^2 \quad \text{subject to} \quad y_i (\vec{w} \cdot \vec{x}^{(i)}) \geq 1 \quad \text{for all } i,$$

and that the margin width of a classifier is $\frac{2}{\|\vec{w}\|}$.

Consider the following training dataset:

i	x_1	x_2	y_i
1	2	-1	+1
2	1	0	+1
3	-1	0	-1
4	-2	3	-1

Assume $b = 0$ throughout. Two candidate weight vectors are proposed:

- $\vec{w}_A = (1, 0)$, so $H_A(\vec{x}) = x_1$
- $\vec{w}_B = (2, 1)$, so $H_B(\vec{x}) = 2x_1 + x_2$

- a) For each candidate, check whether every training point satisfies the Hard-SVM constraint $y_i (\vec{w} \cdot \vec{x}^{(i)}) \geq 1$ for all i . Show your work for each point under each classifier.

Solution: For $\vec{w}_A = (1, 0)$:

$$y_1(\vec{w}_A \cdot \vec{x}^{(1)}) = (+1)(2) = 2 \geq 1 \quad \checkmark$$

$$y_2(\vec{w}_A \cdot \vec{x}^{(2)}) = (+1)(1) = 1 \geq 1 \quad \checkmark$$

$$y_3(\vec{w}_A \cdot \vec{x}^{(3)}) = (-1)(-1) = 1 \geq 1 \quad \checkmark$$

$$y_4(\vec{w}_A \cdot \vec{x}^{(4)}) = (-1)(-2) = 2 \geq 1 \quad \checkmark$$

All constraints satisfied for $\vec{w}_A$.

For $\vec{w}_B = (2, 1)$:

$$y_1(\vec{w}_B \cdot \vec{x}^{(1)}) = (+1)(4 - 1) = 3 \geq 1 \quad \checkmark$$

$$y_2(\vec{w}_B \cdot \vec{x}^{(2)}) = (+1)(2 + 0) = 2 \geq 1 \quad \checkmark$$

$$y_3(\vec{w}_B \cdot \vec{x}^{(3)}) = (-1)(-2 + 0) = 2 \geq 1 \quad \checkmark$$

$$y_4(\vec{w}_B \cdot \vec{x}^{(4)}) = (-1)(-4 + 3) = 1 \geq 1 \quad \checkmark$$

All constraints satisfied for $\vec{w}_B$.

- b) Both classifiers satisfy the Hard-SVM constraints. Compute the margin width for each. Which would the Hard-SVM select, and why?

Solution: For $\vec{w}_A$:

$$\|\vec{w}_A\| = \sqrt{1^2 + 0^2} = 1, \quad \text{margin} = \frac{2}{1} = 2.$$

For $\vec{w}_B$:

$$\|\vec{w}_B\| = \sqrt{2^2 + 1^2} = \sqrt{5}, \quad \text{margin} = \frac{2}{\sqrt{5}}.$$

Since the Hard-SVM minimizes $\|\vec{w}\|^2$, and $\|\vec{w}_A\|^2 = 1 < 5 = \|\vec{w}_B\|^2$, the Hard-SVM selects $\vec{w}_A$, which also has the larger margin width.

- c) Identify all support vectors under $\vec{w}_A$. Recall that a support vector is a training point $\vec{x}^{(i)}$ such that $y_i \vec{w} \cdot \vec{x}^{(i)} = 1$.

Solution: We check which points achieve equality in the constraint under $\vec{w}_A$:

$$\begin{aligned} y_1(\vec{w}_A \cdot \vec{x}^{(1)}) &= 2 \neq 1 \\ y_2(\vec{w}_A \cdot \vec{x}^{(2)}) &= 1 = 1 \quad \checkmark \\ y_3(\vec{w}_A \cdot \vec{x}^{(3)}) &= 1 = 1 \quad \checkmark \\ y_4(\vec{w}_A \cdot \vec{x}^{(4)}) &= 2 \neq 1 \end{aligned}$$

The support vectors are $\vec{x}^{(2)} = (1, 0)$ and $\vec{x}^{(3)} = (-1, 0)$.

- d) A classmate proposes using $\vec{w}_C = (2, 0)$ instead of $\vec{w}_A$, arguing it is just a scaled version and should perform the same.
- (i) Verify that $\vec{w}_C$ satisfies all Hard-SVM constraints.
 - (ii) Compute the margin width under $\vec{w}_C$.
 - (iii) Is your classmate correct? What does this tell you about scaling $\vec{w}$ in the context of Hard-SVM?

Solution: (i) For $\vec{w}_C = (2, 0)$:

$$\begin{aligned} y_1(\vec{w}_C \cdot \vec{x}^{(1)}) &= (+1)(4) = 4 \geq 1 \quad \checkmark \\ y_2(\vec{w}_C \cdot \vec{x}^{(2)}) &= (+1)(2) = 2 \geq 1 \quad \checkmark \\ y_3(\vec{w}_C \cdot \vec{x}^{(3)}) &= (-1)(-2) = 2 \geq 1 \quad \checkmark \\ y_4(\vec{w}_C \cdot \vec{x}^{(4)}) &= (-1)(-4) = 4 \geq 1 \quad \checkmark \end{aligned}$$

(ii)

$$\|\vec{w}_C\| = 2, \quad \text{margin} = \frac{2}{2} = 1.$$

(iii) The classmate is incorrect. While $\vec{w}_C$ and $\vec{w}_A$ define the same decision boundary $H(\vec{x}) = 0$, scaling $\vec{w}$ up increases $\|\vec{w}\|$, which narrows the exclusion zone and reduces the margin. The Hard-SVM penalizes larger $\|\vec{w}\|^2$, so it would always prefer $\vec{w}_A$ over $\vec{w}_C$.

- e) A new training point $\vec{x}^{(5)} = (0.4, 1)$ with label $y_5 = +1$ is added to the dataset.
- (i) Show that this point violates the Hard-SVM constraints under $\vec{w}_A$.

- (ii) Explain briefly why the Hard-SVM optimization problem no longer makes sense with this new point in the dataset.
- (iii) If we instead use a Soft-SVM, what is the minimum slack $\xi_5 \geq 0$ required so that the constraint $y_5(\vec{w}_A \cdot \vec{x}^{(5)}) \geq 1 - \xi_5$ is satisfied?

Solution: (i)

$$y_5(\vec{w}_A \cdot \vec{x}^{(5)}) = (+1)(0.4) = 0.4 < 1.$$

The constraint is violated.

(ii) The dataset is no longer linearly separable, so no $\vec{w}$ can satisfy all Hard-SVM constraints simultaneously. The feasible set of the optimization problem is empty, meaning there is no valid solution.

(iii) We need the smallest $\xi_5 \geq 0$ such that

$$0.4 \geq 1 - \xi_5 \implies \xi_5 \geq 0.6.$$

The minimum required slack is $\xi_5 = 0.6$.

Problem 4.

In this problem, you will explore how the regularization parameter C controls the tradeoff between margin width and classification accuracy in the Soft-SVM. Recall the hinge loss:

$$\ell_{\text{hinge}}(H(\vec{x}), y) = \max\{0, 1 - y H(\vec{x})\},$$

and the regularized SVM risk:

$$R_{\text{svm}}(\vec{w}) = \|\vec{w}\|^2 + \frac{C}{n} \sum_{i=1}^n \max\{0, 1 - y_i \vec{w} \cdot \vec{x}^{(i)}\}.$$

Consider the following training dataset with $n = 4$ points:

i	x_1	x_2	y_i
1	3.0	2	+1
2	0.8	-1	+1
3	-3.0	1	-1
4	-0.8	2	-1

Two candidate predictors (with $b = 0$) are proposed:

- $\vec{w}_A = (1, 0)$, so $H_A(\vec{x}) = x_1$
- $\vec{w}_B = (0.5, 0)$, so $H_B(\vec{x}) = 0.5x_1$

- a) For each predictor, compute $y_i H(\vec{x}^{(i)})$ and the hinge loss ℓ_{hinge} for every training point. Organize your answers in a table with columns: i , $y_i H_A(\vec{x}^{(i)})$, ℓ_{hinge} under $\vec{w}_A$, $y_i H_B(\vec{x}^{(i)})$, ℓ_{hinge} under $\vec{w}_B$.

Solution:

i	$y_i H_A$	$\ell_{\text{hinge}}(\vec{w}_A)$	$y_i H_B$	$\ell_{\text{hinge}}(\vec{w}_B)$
1	3.0	0	1.5	0
2	0.8	0.2	0.4	0.6
3	3.0	0	1.5	0
4	0.8	0.2	0.4	0.6

Note that points 1 and 3 are correctly classified and outside the exclusion zone under both

predictors. Points 2 and 4 are correctly classified but fall inside the exclusion zone ($y_i H < 1$) under both predictors.

- b) Compute $R_{\text{svm}}(\vec{w}_A)$ and $R_{\text{svm}}(\vec{w}_B)$ at $C = 2$. Which predictor does the Soft-SVM prefer at this value of C ?

Solution: The total hinge loss for each predictor:

$$\sum_{i=1}^4 \ell_{\text{hinge}}(\vec{w}_A) = 0 + 0.2 + 0 + 0.2 = 0.4, \quad \sum_{i=1}^4 \ell_{\text{hinge}}(\vec{w}_B) = 0 + 0.6 + 0 + 0.6 = 1.2.$$

With $\|\vec{w}_A\|^2 = 1$ and $\|\vec{w}_B\|^2 = 0.25$:

$$R_{\text{svm}}(\vec{w}_A) = 1 + \frac{2}{4}(0.4) = 1 + 0.2 = 1.2,$$

$$R_{\text{svm}}(\vec{w}_B) = 0.25 + \frac{2}{4}(1.2) = 0.25 + 0.6 = 0.85.$$

At $C = 2$, the Soft-SVM prefers $\vec{w}_B$ since $0.85 < 1.2$.

- c) Now compute $R_{\text{svm}}(\vec{w}_A)$ and $R_{\text{svm}}(\vec{w}_B)$ at $C = 10$. Which predictor does the Soft-SVM prefer now?

Solution:

$$R_{\text{svm}}(\vec{w}_A) = 1 + \frac{10}{4}(0.4) = 1 + 1.0 = 2.0,$$

$$R_{\text{svm}}(\vec{w}_B) = 0.25 + \frac{10}{4}(1.2) = 0.25 + 3.0 = 3.25.$$

At $C = 10$, the Soft-SVM prefers $\vec{w}_A$ since $2.0 < 3.25$.

- d) Find the exact value of C at which both predictors have equal regularized risk. *Hint: express R_{svm} as a linear function of C for each predictor and set them equal.*

Solution: Writing each risk as a function of C :

$$R_{\text{svm}}(\vec{w}_A) = 1 + \frac{C}{4}(0.4) = 1 + 0.1C,$$

$$R_{\text{svm}}(\vec{w}_B) = 0.25 + \frac{C}{4}(1.2) = 0.25 + 0.3C.$$

Setting them equal:

$$1 + 0.1C = 0.25 + 0.3C \implies 0.75 = 0.2C \implies C = 3.75.$$

- e) In one or two sentences, explain intuitively why the preferred predictor changes as C increases. Your answer should reference what $\|\vec{w}\|^2$ and the hinge loss terms each represent in the optimization.

Solution: $\|\vec{w}\|^2$ controls the margin width — smaller $\|\vec{w}\|^2$ gives a wider margin — while the hinge loss terms penalize points that are misclassified or too close to the decision boundary. At small C , the margin term dominates and $\vec{w}_B$ is preferred because its smaller $\|\vec{w}\|^2$ more than compensates for its larger hinge loss. As C grows, the penalty on hinge loss becomes dominant, and $\vec{w}_A$ is preferred because it keeps more points safely outside the exclusion zone.

- f) As $C \rightarrow \infty$, what does the Soft-SVM approach? What condition on the data is required for this limit to be well-defined, and does this dataset satisfy it? Justify your answer.

Solution: As $C \rightarrow \infty$, any nonzero hinge loss is penalized infinitely heavily, so the Soft-SVM approaches the Hard-SVM, which enforces $y_i \vec{w} \cdot \vec{x}^{(i)} \geq 1$ for all i with no slack allowed. This limit is only well-defined when the data are linearly separable, since otherwise no feasible solution exists. This dataset is linearly separable: the decision boundary $x_1 = 0$ perfectly separates the positive points ($x_1 > 0$) from the negative points ($x_1 < 0$), so the limit is well-defined here.